Nayak *et al. Journal of Big Data* (2025) 12:138
https://doi.org/10.1186/s40537-025-01196-5

Journal of Big Data

RESEARCH Open Access

Meta-transformer: leveraging metaheuristic algorithms for agricultural commodity price forecasting

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Abstract

Predicting agricultural commodity prices is inherently complex due to factors such as perishability, seasonality, and market volatility. To address these challenges, this study proposes a novel framework that combines Transformer models with Metaheuristic Algorithms (MHAs), including the Grey Wolf Optimizer (GWO), Whale Optimization Algorithm (WOA), and Particle Swarm Optimization (PSO) to enhance agricultural price forecasting accuracy. While Transformer architectures are known for their powerful time series modeling capabilities, their performance is highly sensitive to hyperparameter selection, especially in contexts with limited or noisy data. The novelty of this research lies in the automated and adaptive tuning of these hyperparameters using MHAs, enabling improved generalization, faster convergence, and enhanced predictive accuracy. By integrating MHAs, known for their fast convergence and global search efficiency, the proposed models, Transformer-PSO, Transformer-GWO, and Transformer-WOA offer enhanced training efficiency and improved forecasting accuracy. This hybrid modeling approach is applied to predict weekly prices of potatoes in key Northern Indian markets. Results demonstrate that the Transformer-GWO and Transformer-WOA models outperform conventional models such as GARCH by 70–90% across standard evaluation metrics like Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). By bridging state-of-the-art deep learning architectures with robust optimization strategies, this study contributes a scalable and interpretable solution for agricultural price forecasting. The findings have significant implications for policymakers, market regulators, and farmers by supporting timely interventions, improving market transparency, and enabling data-driven decision-making.

Keywords: Agricultural price forecasting, Deep learning, Metaheuristic algorithms, Optimization, Transformer

Introduction

Agriculture is the backbone of India's economy, which supports the livelihoods of nearly half of its population and plays a crucial role in food security, employment generation and economic growth [1]. However, the agricultural sector is highly susceptible to market fluctuations, climate variability and policy changes, which often result in volatile

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commodity prices [2]. Accurate forecasting of agricultural commodity prices is essential for ensuring market stability, supporting farmers' decision-making and guiding policymakers in developing effective interventions. Among staple crops, the potato holds significant importance as the third most consumed food crop globally after rice and wheat. Introduced to India in the seventeenth century, it has become an essential part of the Indian diet and farming systems [3]. In 2023, India's potato production reached 59.74 million metric tons, with Uttar Pradesh making the largest contribution [4]. Potatoes, known for their high nutritional value, serve as both a staple food and a source of income for farmers. They cover nearly 20.79% of India's total vegetable cultivation area [5]. The global potato market is projected to reach $137.46 billion by 2029, driven by the growing demand from the processing industry [6]. Alongside the growing importance of potatoes, price forecasting has emerged as a rapidly expanding area of research, particularly in the agricultural sector. The relevance of agricultural commodity price forecasting has grown significantly, as timely and accurate predictions are crucial for farmers, consumers, and policymakers alike. However, effective forecasting remains challenging due to inherent complexities such as seasonality, fluctuating demand, climate variability, market imperfections, and the impact of globalization. Recent literature highlights a wide range of efforts to improve forecasting accuracy using various time series (TS) models, emphasizing the need to address these challenges in the dynamic and evolving landscape of agricultural markets.

Traditional statistical models in time series forecasting encounter limitations as they assume a predetermined functional form, hindering their ability to capture intricate patterns and nonlinear relationships in the data [7, 8]. These models heavily rely on stringent assumptions about data distribution, potentially resulting in less accurate forecasts in real-world scenarios. While machine learning (ML) models have proven effective in various applications, they may face challenges in capturing long-term dependencies and require manual feature extraction [9–11].

However, recent advances in deep learning (DL) provide a promising solution to these challenges. DL models excel at automatically learning complex and non-linear relationships, making them well-suited for handling large datasets and enhancing forecasting accuracy [12]. DL models operate efficiently with TS data, offering scalable solutions that yield reliable predictions [13–15]. Their unique capabilities in capturing complex patterns differentiate DL from statistical and ML techniques, empowering them to model intricate relationships between inputs and outputs effectively. However, in conventional DL architectures such as Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU), the input data are sequentially processed, which is time-consuming, also, RNNs may face vanishing/exploding gradient issues, LSTMs demand more computational resources and GRUs may struggle with very long-term dependencies. The introduction of the transformer architecture with its innovative attention mechanism has significantly transformed multiple fields [16]. Although transformer-based models have been successfully applied across various domains, their use in forecasting agricultural prices remains largely unexplored [17, 18]. Despite these strengths, achieving optimal performance with DL models demands well-tuned hyperparameters, prompting the exploration of suitable optimization techniques[19]. The performance of the Transformer model, renowned for its versatility in various

tasks, is heavily dependent on several hyperparameters. Key parameters such as the number of neurons, choice of activation function, optimizer type, learning rate, learning rate decay, batch size, number of training epochs, head size, number of attention heads, feed-forward layer dimensions and the number of Transformer blocks all critically influence the model's forecasting accuracy. Adjusting these parameters is essential for optimizing the model's effectiveness in specific applications. In recent studies on TS prediction tasks, researchers have explored innovative approaches to enhance the performance of various models. Several studies have explored the significance of DL models in the domain of financial TS price forecasting. Roy Choudhury et al. [20] introduced an autoencoder-based approach utilizing LSTM networks for stock price forecasting. Tripathi and Sharma [21] found that deep neural networks (DNNs) outperformed LSTM and CNN-LSTM models in predicting Bitcoin prices using technical indicators. Durairaj and Mohan [22] proposed a CNN-based model for price forecasting incorporating chaos theory, 1D CNN and polynomial regression. Paquet and Soleymani [23] introduced Quantum Leap, a DQNN model for price forecasting. Avinash et al. [8] put forth Hidden Markov (HM) guided DL models for forecasting agricultural commodity prices. Zhang et al. [24] reviewed the importance of DL models for price forecasting in financial TS. Furthermore, researchers have explored the integration of metaheuristic algorithms (MHAs) to enhance the performance of DL models. Wang et al. [25–27] focused on optimizing Echo State Networks (ESN) using the Grey Wolf Algorithm (GWO). Louis et al. [28] developed an Artificial Neural Network (ANN) model for water level prediction, employing the Whale Optimization Algorithm (WOA). Diop et al. [29] hybridized a Multilayer Perceptron (MLP) model with WOA for annual rainfall prediction. Fu and Li [30] proposed a hybrid model, VMD-WOA-LSTM, for accurate evapotranspiration estimation. Cinar and Natarajan [31] applied an ANN optimized by GWO for hourly wind speed prediction. Kilinc and Yurtsever [32] introduced a hybrid GWO-GRU model for streamflow forecasting, while [33] improved LSTM-based wind speed prediction using the WOA. These studies collectively demonstrate the versatility and effectiveness of combining advanced optimization algorithms with various models for improved TS forecasting in diverse domains [34–36].

Traditional time series forecasting models exhibit notable limitations in handling the complex dynamics of agricultural price series. Statistical models like GARCH are built on linear assumptions and often fall short in capturing nonlinearities, regime shifts and multi-scale variations typical in real-world agricultural markets. Artificial Neural Networks (ANNs), while more flexible, require manual feature engineering and often lack memory to model sequential dependencies, making them prone to overfitting when data is noisy or limited. LSTM networks, though designed for sequential data, are computationally intensive and sensitive to hyperparameter settings. Tuning parameters such as the number of layers, units, dropout rates, and learning rate can be challenging, and improper configurations may lead to slow convergence or suboptimal performance. These challenges collectively underline the need for more robust, scalable and adaptive models, motivating our proposed approach that leverages the Transformer architecture enhanced by metaheuristic optimization algorithms.

Remarkably, despite their critical importance, the integration of Transformer DL architectures with MHAs [37] for modeling agricultural commodity prices has been

conspicuously absent from the existing literature. This synergy presents a unique and untapped opportunity to enhance forecasting models' predictive performance while simultaneously addressing hyperparameter tuning challenges. Such an integration could lead to more accurate and efficient predictions of agricultural commodity prices. This research gap highlights the need for exploration and experimentation in integrating the Transformer model with MHAs for improved forecasting outcomes in the agricultural domain. Therefore, for the first time, this study bridges that gap by introducing a novel framework that unites the advanced computational capabilities of Transformer architectures with the robust optimization strengths of MHAs. The study presents novel contributions that encompass both technological innovation and practical application in the agricultural domain as follows:

- Development of specific Meta-Transformers
- Hyperparameter optimization via MHAs
- Empirical evaluation across multiple markets
- Contribution to methodological advances in agricultural forecasting

The novelty of this study lies in the integration of Transformer architectures with Metaheuristic Algorithms (MHAs) such as Grey Wolf Optimizer, Whale Optimization Algorithm, and Particle Swarm Optimization. While Transformer models are well-known for their ability to capture complex temporal dependencies and allow parallelized training, their performance is highly dependent on the careful tuning of numerous hyperparameters. Traditional tuning methods such as grid or random search are computationally expensive and suboptimal in high-dimensional search spaces. MHAs offer a more efficient and adaptive mechanism for exploring these spaces, enabling faster convergence and better generalization. This synergy allows the proposed Meta-Transformer models to achieve superior forecasting accuracy with reduced computational overhead, distinguishing them from standard Transformer implementations and existing deep learning approaches in agricultural forecasting.

The significance of accurate agricultural price forecasting extends beyond farmers' income stability. It underpins strategic decision-making in supply chain logistics, policy formulation, and national food security. Reliable forecasts help to reduce market inefficiencies, minimize post-harvest losses, and inform stakeholders about optimal storage, transport, and procurement strategies. For policymakers, price prediction models support evidence-based interventions such as Minimum Support Prices (MSP), subsidies, and import–export controls, ultimately contributing to economic resilience and inflation control. In the private sector, agri-tech platforms and retailers can use these insights for real-time market optimization and contract planning. More importantly, existing Transformer models often rely on manual or basic hyperparameter tuning, which hampers generalization and efficiency. Our work is the first to introduce a Meta-Transformer framework that integrates advanced metaheuristic optimization algorithms (GWO, WOA, PSO) with Transformers for this domain. Unlike prior approaches, our models adaptively optimize hyperparameters, leading to faster convergence and more accurate forecasts. We further validate our approach on real-world datasets across five diverse markets, showcasing robustness under varying volatility and data conditions. These

contributions establish a strong foundation for scalable and intelligent agri-market forecasting systems.

Materials and methods

Transformer

The Transformer model, originally introduced by Vaswani et al. [16], has proven to be a potent architecture (see Fig. 1) in natural language processing tasks, with its application later extended to TS forecasting [38–40]. The model utilizes a self-attention mechanism to effectively capture the significance of different elements in input sequences, thereby enhancing prediction accuracy. Its architecture features an encoder-decoder structure incorporating input embedding and positional encoding for preserving sequence information. Stacked layers in the encoder consist of multi-head self-attention and feedforward components, while the decoder mirrors this structure with additional encoder-decoder attention. The self-attention mechanism employs linear combinations of query, key and value vectors, and multi-head attention enables parallel processing.

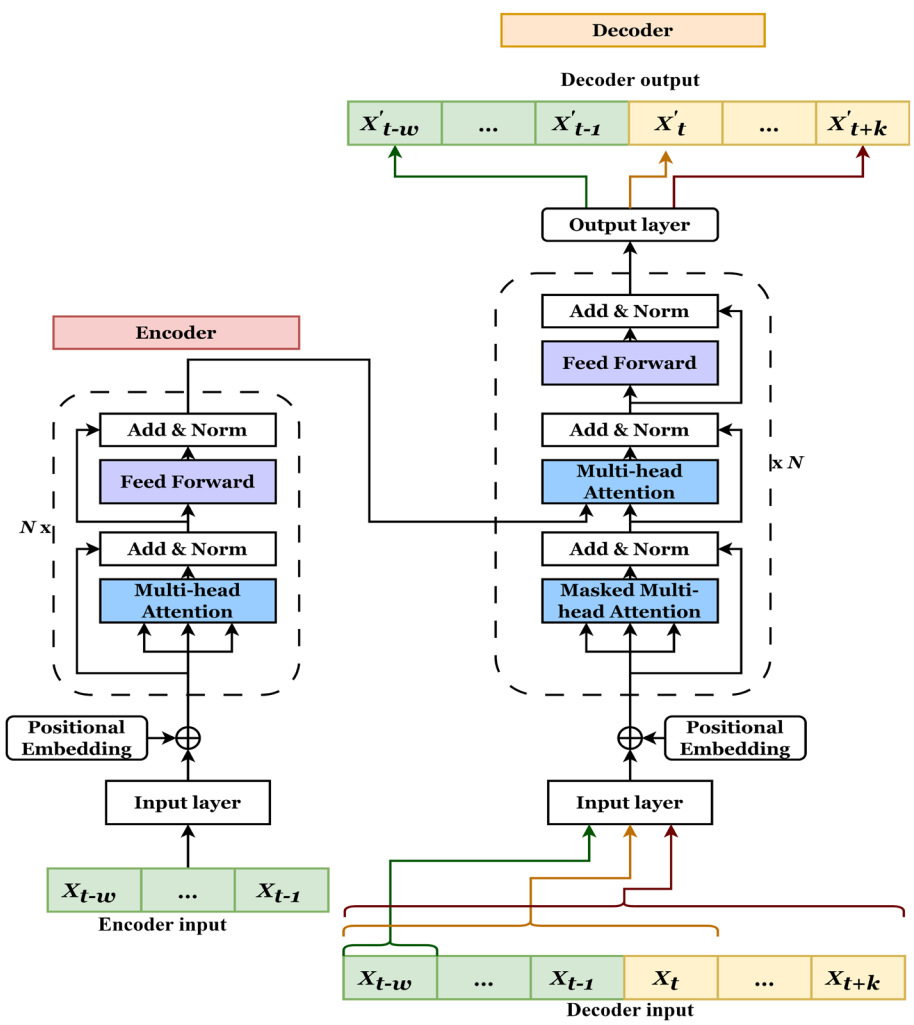

Fig. 1 Architecture of Transformer

Masking during training prevents the model from using future data, and stability is enhanced through residual connections. The Transformer's versatility and parallel processing capabilities position it as a robust choice for various nonlinear dynamical systems and time series data, showcasing superior efficiency and effectiveness compared to traditional models like RNNs, LSTMs and GRUs [41, 42].

Metaheuristic algorithms (MHAs)

Optimization techniques, vital for refining TS models in forecasting, often face challenges with local optima convergence in deterministic local search algorithms. In contrast, MHA, acknowledged for their suitability in global optimization, provides optimal solutions. MHA operates on a problem-independent, heuristic algorithmic framework, aiming to deliver "good enough" solutions within limited computing time. Noteworthy features include adaptability to real-time problems, superior solution quality in less time and a non-deterministic nature preventing premature convergence [43]. The literature of the past decades reflects a surge in MHA, classified into "classical" (e.g., Genetic Algorithm (GA), Particle Swarm Optimization (PSO) etc.) and "new generation" (e.g., Grey Wolf Optimizer (GWO), Whale Optimization Algorithm (WOA) etc.) algorithms [44]. This evolution addresses challenges associated with local optima, computing time and problem adaptability. Traditional optimization techniques exhibit limitations, including premature convergence, entrapment in local optima, high computational complexity and poor performance in high-dimensional problems [45]. The new generation MHAs overcome these limitations. They have a remarkable capacity to prevent premature convergence and their primary goal is to find the optimal solution with minimal computing complexity in a fair amount of time [43].

The GWO and WOA are specific types of MHAs that stand out among various optimization techniques due to their unique advantages. Notably, both GWO and WOA exhibit simplicity, faster convergence, fewer parameters and robustness. Despite the multitude of available MHAs, these characteristics make GWO and WOA appealing choices for optimizing DL models. These algorithms aid in obtaining optimal hyperparameters and the resulting best model is then built and trained using these optimized hyperparameters. The optimization process is driven by the search conducted by these algorithms in the hyperparameter space. By integrating MHAs with DL models, we can efficiently navigate expansive solution spaces, discover optimal configurations and enhance forecasting accuracy. The related techniques are elaborated in detail further.

Grey wolf optimization (GWO):

GWO draws inspiration from the social structure and hunting practices of wolves [46]. The algorithm starts with a swarm of candidate solutions, where each candidate represents a potential solution to the optimization problem. The algorithm iteratively updates candidate solution positions based on the social hierarchy and hunting behaviour of grey wolves, showcasing its effectiveness in achieving optimal solutions. Drawing parallels from the alpha–beta-omega hierarchy observed in wolf packs, GWO leverages a dynamic approach to global optimization [47, 48]. Alphas, representing leaders, guide the optimization process, betas assist in exploration and decision-making and omegas contribute to diversifying the search space. Subordinate roles, mirroring various pack

functions, aid in refining solutions, monitoring boundaries and overall optimization (as shown in Fig. 2). The GWO algorithm, inspired by the organizational principles of wolf packs, exhibits promising outcomes, particularly in the efficient exploration of hyperparameter spaces for DL models.

The mathematical model of the GWO algorithm is based on the following equations:

1. Position update equations

$$X_{i,t+1} = X_{p,t} + A_1.D_1 \tag{1}$$

$$X_{i,t+1} = X_{\alpha,t} - A_2.D_2 \tag{2}$$

where, $X_{i,t+1}$ is the position of the ith candidate solution at iteration $t+1$. $X_{p,t}$ is the position of the dominant grey wolf (alpha wolf) at iteration t. $X_{\alpha,t}$ is the position of the alpha wolf at iteration t. A_1 and A_2 are coefficient vectors. D_1 and D_2 are random vectors.

2. Coefficient vector equations

The coefficient A_1 and A_2 are calculated using the following equations:

$$A_1 = 2\alpha.r_1 - a \tag{3}$$

$$A_2 = 2\alpha.r_2 - a \tag{4}$$

where, a is a linearly decreasing parameter that starts from 2 and decreases to 0 as the algorithm progresses. r_1 and r_2 are random vectors in the range $[0, 1]$.

3. Random vector equations

The random vectors D_1 and D_2 are calculated using the following equations:

$$D_1 = \left|C_1.X_{\alpha,t} - X_{i,t}\right| \tag{5}$$

$$D_2 = \left|C_2.X_{\beta,t} - X_{i,t}\right| \tag{6}$$

where, C_1 and C_2 are random vectors in the range $[-1, 1]$. $X_{\alpha,t}$ and $X_{\beta,t}$ are the positions of the alpha wolf and beta wolf (second dominant wolf) at iteration t.

Whale optimization algorithm (WOA):

WOA is inspired by the hunting strategies of whales [49]. The algorithm maintains a population of candidate solutions, or whales, where each whale represents a potential solution to the optimization problem. Guided by a humpback whale leader steering the group toward a food source and a sperm whale aiding in finding the optimal food source, WOA showcases a collaborative approach akin to the social behaviours observed in whale populations. Whales exhibit intriguing characteristics, have spindle cells in their brains responsible for judgment and emotions similar to humans and display unique

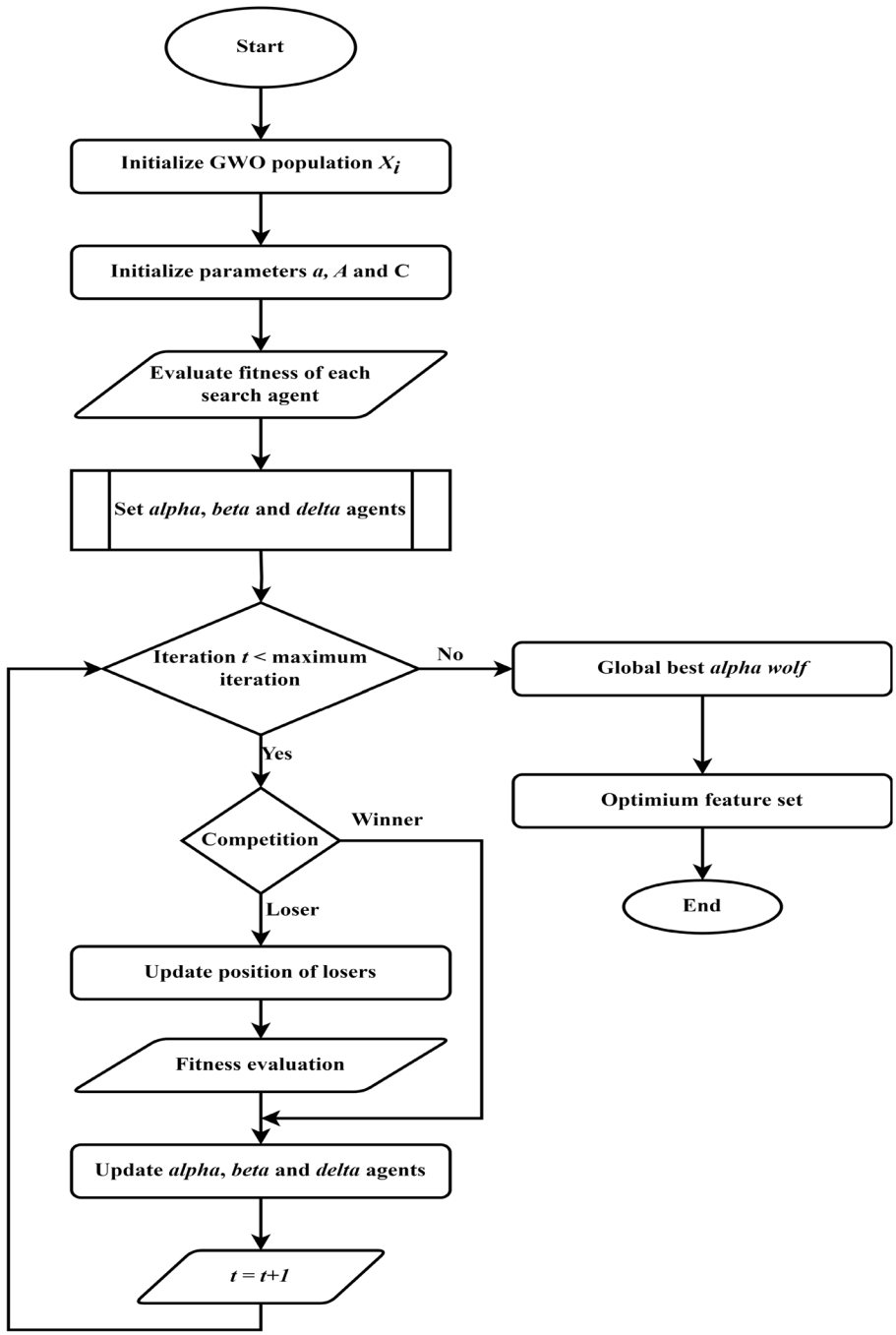

Fig. 2 Flowchart for GWO algorithm

social behaviours, including dialect development among killer whales. The distinctive bubble-net feeding method employed by humpback whales for hunting krill and small fish further highlights their exceptional adaptability and intelligence [50, 51]. The flowchart of WOA algorithm is displayed in Fig. 3.

The mathematical model of WOA is grounded in two main equations representing the encircling prey and bubble-net attacking method:

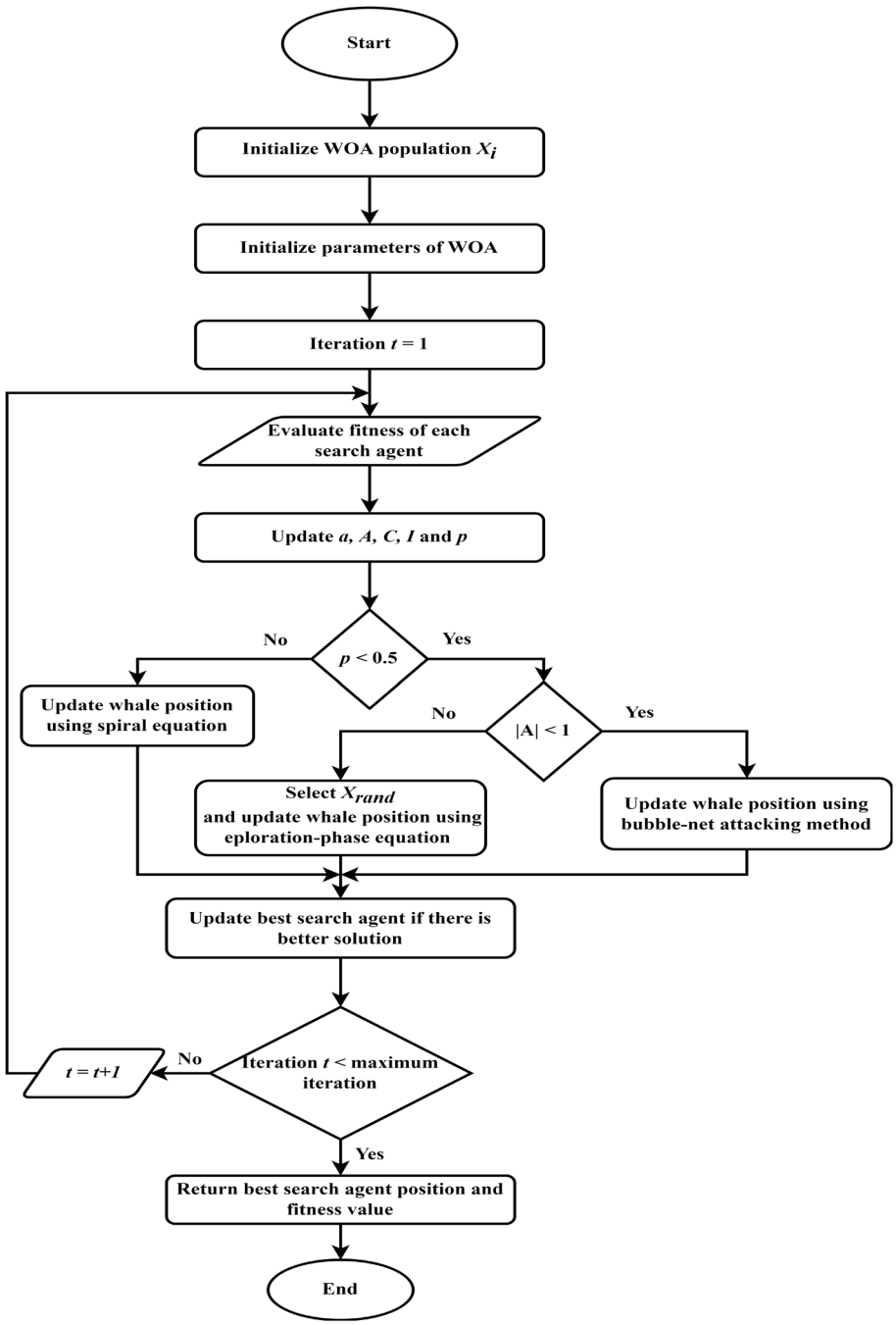

Fig. 3 Flowchart for WOA algorithm

1. Encircling prey

$$X_i(t+1) = X_{best}(t) - A.D_i(t) \tag{7}$$

where, $X_i(t+1)$ is the position of whale i at iteration $(t+1)$, $X_{best}(t)$ is the position of the best whale at iteration t, A and $D_i(t)$ are coefficient vectors calculated as follows:

$$A = 2a.r_1 - a \tag{8}$$

$$D_i(t) = |C.X_{best}(t) - X_i(t)| \tag{9}$$

where, a is a linearly decreasing coefficient from (2 to 0), random vectors (r_1 and r_2) in the range $[0, 1]$, C is a coefficient vector calculated a:

$$C = 2.r_2 \tag{10}$$

2. Bubble-net attacking methodwhere, l is a random number in the range $[-1, 1]$.

$$X_i(t+1) = X_{best}(t) - A.D_i(t).e^{bl}.cos(2\pi l) \tag{11}$$

The GWO and the WOA stand out as powerful solutions for diverse optimization challenges. GWO, a population-based algorithm, excels in exploring various regions of the search space simultaneously, enhancing its capabilities for comprehensive solution discovery. Its simplicity of implementation and minimal parameter tuning requirements contribute to its broad appeal. Demonstrating efficacy across multimodal problems, constrained scenarios and real-world applications, GWO finds successful applications in engineering design, financial optimization, machine learning, data mining, image processing and robotics. Similarly, WOA, operating as a population-based algorithm, distinguishes itself by concurrently exploring diverse regions of the search space, enriching its capacity for solution discovery. Noteworthy for its efficacy, simplicity of implementation and minimalistic parameter tuning requirements, WOA has demonstrated success in addressing multimodal problems, navigating through constrained scenarios and providing effective solutions in real-world applications. Its versatility is evident across engineering design, financial optimization, machine learning, data mining, image processing and robotics, making WOA a valuable asset in tackling intricate optimization tasks across various domains. Both GWO and WOA showcase adaptability and reliability, underscoring their significance in addressing complex optimization challenges [52].

Particle swarm optimization (PSO)

PSO, simulating the social behaviour of birds and fish [53]. It involves a population of particles navigating through the search space to discover the optimal solution. Each particle dynamically adjusts its position based on both its individual experience and the collective experience of its neighbours. The algorithm's design mirrors the coordinated movements observed in bird flocks, contributing to its efficiency in exploring solution spaces. Each particle in the PSO algorithm represents a potential solution to the optimization problem, characterized by two vectors, position and velocity, with the former encapsulating variable values. The dual-vector system enables particles to efficiently explore n-dimensional search spaces, where n corresponds to the number of variables in the problem. The flowchart of the PSO algorithm is shown in Fig. 4.

The mathematical model of the PSO algorithm:

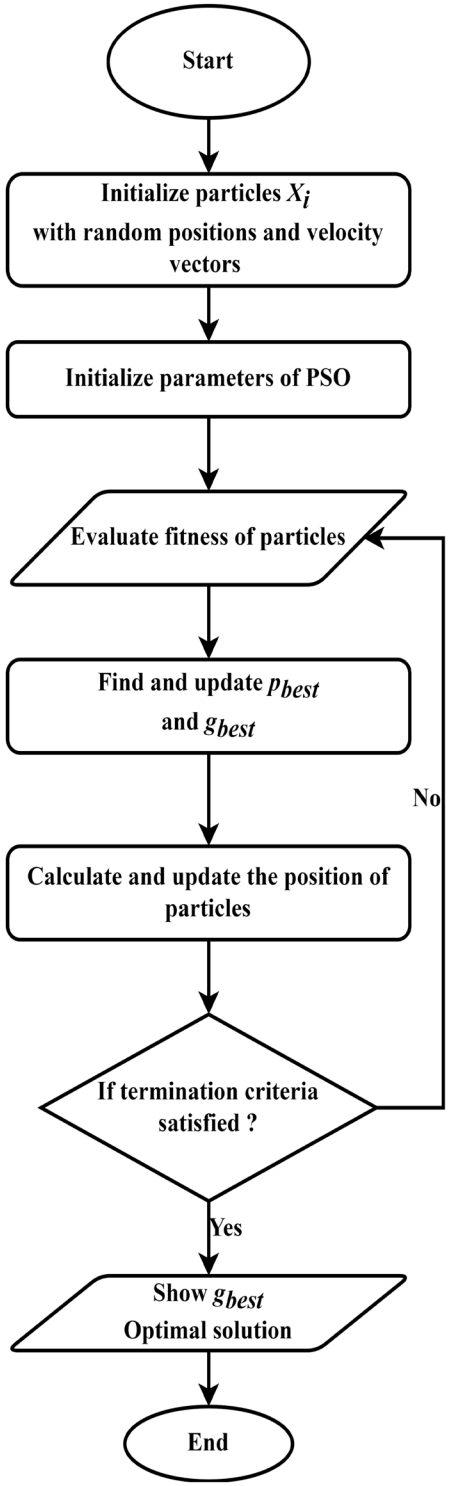

Fig. 4 Flowchart for PSO algorithm

1. **Initialization:** Initialize the population of particles with random positions X_i and velocities V_i in the search space.

2. **Update velocities and positions:** Update the velocities and positions of each particle using the following equations:

$$V_i(t+1) = \sqsupseteq.V_i(t) + c_1.r_1.(Pbest_i - X_i(t)) + c_2.r_2.(Gbest - X_i(t)) \tag{12}$$

$$X_i(t+1) = X_i(t) + V_i(t+1) \tag{13}$$

where, $V_i(t)$ is the velocity of the particle i at time t. $X_i(t)$ is the position of particle i at time t. $Pbest_i$ is the personal best position of particle i. $Gbest$ is the global best position. $\sqsupseteq$ is the inertia weight. c_1 and c_2 are acceleration coefficients. r_1 and r_2 are random values in the range [0, 1].

3. **Update personal best (*Pbest*) and Global best (*Gbest*):** Update the personal best position $Pbest_i$ for each particle and the global best position $Gbest$ based on the fitness of the particles.
4. **Termination:** Repeat the velocity and position updates until a termination condition is met (For example, maximum iterations).

Assessment criteria

a) Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \widehat{y}_i\right)^2} \tag{14}$$

b) Mean Absolute Error (MAE)

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \widehat{y}_i\right| \tag{15}$$

where, n is the number of observations, y_i is the actual value and $\widehat{y}_i$ is the predicted value.

The framework of the proposed meta-transformer model

Combining the Transformer architecture with MHAs presents a scientifically robust methodology that provides distinct advantages in terms of architectural enhancements. The proposed framework is depicted in Fig. 5. The Transformer's inherent capacity for capturing long-range dependencies and contextual information is complemented by the optimization strengths of MHAs. The self-attention mechanism of the Transformer allows for efficient feature representation, while the incorporation of MHAs introduces benefits such as accelerated convergence, superior parameter tuning and effective exploration of a vast solution space. This symbiotic integration empowers the architecture with a heightened ability to adapt to complex data patterns and uncertainties, resulting in a scientifically optimized model capable of achieving superior performance.

The Transformer model's crucial hyperparameters guide the model's architecture and training dynamics. The 'Head size parameter influences the complexity of relationships captured by attention heads, crucial for identifying temporal patterns. The 'Number of

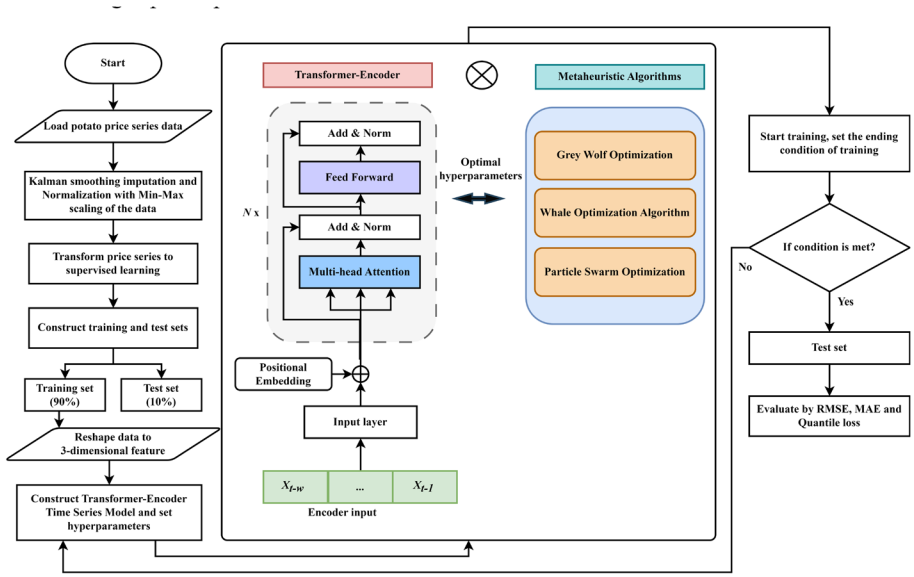

Fig. 5 Flowchart of the proposed Meta-Transformer model

heads' hyperparameter determines the number of attention heads, enabling the model to simultaneously focus on diverse temporal aspects. The 'feed-forward dimension' parameter impacts the feedforward layer's dimensionality, affecting the model's ability to capture non-linear temporal dependencies. The 'Number of Transformer blocks' parameter influences the depth of the model, allowing it to learn hierarchical temporal features. The 'Number of neurons' hyperparameter specifies the size of the feedforward layer, influencing the model's capacity to discern intricate temporal patterns. Activation functions, optimizers, learning rate, decay rate, batch size and epochs play pivotal roles in shaping the model's ability to forecast time series data. The hyperparameter tuning process involves selecting values within the specified ranges to optimize the model's forecasting performance, balancing complexity and computational efficiency for time-sensitive applications.

Proposed algorithm for meta-transformer model

Objective

Optimize the hyperparameters of a Transformer model using the MHA to enhance forecasting performance.

1. **Input:** Historical price data series $P = \{p_1, p_2, \ldots, p_T\}$.
2. **Output:** Forecasted values $\widehat{P} = \left\{\widehat{p}_{T+1}, \widehat{p}_{T+2}, \ldots, \widehat{p}_{T+h}\right\}$ for h future time steps.
3. **Pre-processing Stage:**
- Handle missing values through Kalman smoothing imputation.
- Normalize the data: $P_{norm} = \frac{P - min(P)}{max(P) - min(P)}$.
4. **Preparation of data:**
- Splitting the data: Divide R_{norm} into train (P_{train}) and test (P_{test}) sets.
- Supervised learning transformation:

- Input: Create input sequences X consisting of windows of consecutive values from P_{train}, represented as $\{x_{t-1}, x_{t-2}, \ldots, x_{t-w}\}$.
- Output: Form corresponding output sequences Y that contain the future values, denoted as $\{x_t', x_{t+1}', \ldots, x_{t+k}'\}$.

5. **Define Model hyperparameters:**
- Initialize the MHA with the specified hyperparameter ranges.

Hyperparameters	Range
Head size: *'head_size'*	(16–128)
Number of attention heads: *'num_heads'*	(1–4)
Feed-Forward dimension: *'d_ff'*	(16–128)
Number of Transformer blocks: *'num_transformer_blocks'*	(1–4)
Number of neurons: *'num_neurons'*	(16–128)
Activation function: *'act_fct'*	(*'relu'*, *'tanh'*, *'sigmoid'*)
Optimizer:*'optimizer'*.	(*'adam'*, *'sgd'*, *'rmsprop'*)
Learning rate: *'lr'*	(0.0001–0.01)
Decay rate: *'decay'*	(0.0001–0.01)
Batch size:*'batch_size'*.	(16–128)
Number of epochs: *'epochs'*	(10–100)

- Randomly initialize a population of candidate solutions (hyperparameter combinations).
- Evaluate each candidate solution by training and validating the Transformer model with the given hyperparameters.
- Update the positions based on their fitness.
- Repeat steps 6–8 until the termination criterion (e.g., maximum number of iterations) is met.

6. **Determining Optimal Hyperparameters:**
- The best hyperparameter combination found during the optimization process is considered the optimal set of hyperparameters.

7. **Training the Transformer model with Optimal Hyperparameters:**
- Train the Transformer model using the optimal hyperparameters obtained.
- Evaluate the performance of the model on a holdout test set.

8. **Model Forecasting:**
- Forecast using P_{test}: $\widehat{P} = Model(P_{test})$.

9. **Post-Processing:**
- Inverse normalization: $\widehat{P}_{original} = P_{norm} \times (max(P) - min(P)) + min(P)$.

10. **Performance Evaluation:**
- Calculate various evaluation metrics such as RMSE, MAE to assess the model's performance.

Mathematically, the hyperparameter tuning problem can be framed as an optimization problem:

$$\theta^* = \underset{\theta \in \Theta}{\operatorname{argmin}} L\left(f_\theta(X), Y\right)$$

where,

θ represents the set of hyperparameters,

L denotes the loss function,

$f_\theta(X)$ is the Transformer model output given input X,

Y is the ground truth.

Traditional techniques like grid search and random search become inefficient in high-dimensional or non-convex search spaces. In contrast, MHAs (GWO, WOA and PSO) offer intelligent and adaptive mechanisms for exploring complex optimization landscapes, enabling efficient discovery of near-optimal hyperparameter configurations.

Optimization techniques enhance the accuracy of time series forecasting models by identifying optimal model parameters. Traditional methods like PSO and GA often face challenges such as premature convergence, high complexity, and poor scalability. In contrast, GWO and WOA effectively address these issues by offering adaptive, efficient, and robust optimization with minimal computational cost. This study employs GWO and WOA-known for their simplicity, faster convergence, fewer parameters and strong performance, for hyperparameter tuning of deep learning models.

PSO is generally faster in early convergence, making it suitable for time-sensitive tuning but may sacrifice diversity. GWO offers a balance between performance and computational cost, making it highly stable for repeated experiments. WOA, while strong in exploration, may require more iterations due to its oscillatory behaviour and complex update mechanisms.

Results and discussion

Implementing the aforementioned forecasting models comprises five essential stages: Data description, Data preprocessing, Model compilation, Model training, validation and evaluation. The subsequent section elaborates in detail.

Data description

In this study, weekly Potato price series (Rs. /Quintal) from five key markets in India's Northern zone were obtained from 'AgMarknet' (https://agmarknet.gov.in) and are listed in Table 1. The time series plots of different markets are presented in (Appendix: Fig). The descriptive statistics are reported in Table 2. The descriptive statistics reveal significant variations in price distributions across the five markets: Dehradun,

Table 1 Data description of potato price series of different markets

Markets	Number of observations	Date (From–To)	Frequency of the dataset
Dehradun (Uttarakhand)	1093	February, 2003 to January, 2024	Weekly
Farrukhabad (Uttar Pradesh)	1077	January, 2003 to August, 2023	Weekly
Ludhiana (Punjab)	944	December, 2005 to January, 2024	Weekly
Pipli (Haryana)	734	January, 2010 to January, 2024	Weekly
Una (Himachal Pradesh)	879	March, 2007 to January, 2024	Weekly

Table 2 Descriptive statistics of different markets considered under study

Descriptive Statistics	Markets				
	Dehradun	Farrukhabad	Ludhiana	Pipli	Una
Minimum	50	10.63	150.00	150.00	208.33
Mean	646.75	98.85	624.76	981.80	896.03
Maximum	3200	438.80	3100.00	4150.02	2360.00
Standard Deviation	425.07	65.39	335.52	796.62	432.67
Coefficient of Variation (%)	65.72	66.14	53.70	81.13	48.28
Skewness	2.20	1.64	1.42	1.98	0.86
Kurtosis	6.89	4.02	3.85	4.23	0.34
Jarque–Bera test	3024.07 (< 0.0001)	1203.44 (< 0.0001)	893.92 (< 0.0001)	1017.11 (< 0.0001)	113.88 (< 0.0001)
Shapiro–Wilk's test	0.80 (< 0.0001)	0.86 (< 0.0001)	0.89 (< 0.0001)	0.78 (< 0.0001)	0.93 (< 0.0001)

Note: The values in the parenthesis indicate p-value

Farrukhabad, Ludhiana, Pipli and Una. For instance, Dehradun had a minimum price of Rs. 50 per quintal, while Pipli reached as high as Rs. 4150.02 per quintal. The mean prices ranged from Rs. 138.65 per quintal in Dehradun to Rs. 624.76 per quintal in Ludhiana. Standard deviations, indicating volatility, varied substantially, particularly in Pipli and Una. The coefficient of variation highlighted the relative variability, with Pipli exhibiting the highest at 81.13 percent. Skewness values suggested non-normal distributions across all markets, with Dehradun showing the most positively skewed distribution. Kurtosis values indicated varying degrees of tail weight, with Dehradun having the heaviest tails again. Results from the Jarque–Bera test and the Shapiro–Wilk's test confirmed the departure from normality in all markets.

Data pre-processing

Data pre-processing is a critical step in ensuring the quality, reliability and usability of datasets for effective analysis and model development. For this study, each time series dataset was divided into training, validation and testing sets to ensure rigorous model evaluation. The training set consisted of 90% of the initial weekly observations, used for model construction and in-sample prediction. From the training set, 20% of the data was further allocated as a validation set to fine-tune the model during the training phase, optimizing hyperparameters and preventing overfitting. The remaining 10% of the dataset was reserved as the testing set for post-sample predictions and final model evaluation.

To address missing or noisy data, Kalman smoothing imputation was employed. This method, known for its effectiveness in time series datasets, ensured that imputed values accurately reflected underlying trends, minimizing distortions in the analysis. Following imputation, the datasets were normalized to a [0, 1] range using Min–Max normalization. Normalization ensured uniform scaling across all features, enabling efficient model convergence and avoiding dominance by any single feature during training.

The validation set played a pivotal role in assessing the model's performance on unseen data during training, providing feedback for iterative improvements. This additional layer of evaluation ensured the model generalized effectively to new data, reducing overfitting and improving robustness. Together, these pre-processing steps—including strategic dataset splitting, imputation and normalization—formed a comprehensive pipeline that enhanced model performance and adhered to best practices in time series analysis and machine learning [54].

Model building

Construct the Transformer model using the optimal hyperparameters obtained from MHAs and evaluate on the unseen data.

Hyperparameter tuning

The effectiveness of DL models is contingent upon the optimal tuning of hyperparameters [55]. For Transformer models, crucial hyperparameters include the number of neurons, activation function, optimizer, learning rate, decay, batch size, number of epochs, head size, number of heads, feed-forward dimension and the number of Transformer blocks. These parameters significantly influence the model's complexity, pattern recognition capabilities, learning efficiency, convergence behaviour, generalization potential and ability to capture sequential dependencies. Effective tuning is paramount for maximizing accuracy and robustness. In our study, we conducted an extensive exploration of hyperparameter values within the following ranges: head size (16–128), number of heads (1–4), feed-forward dimension (16–128), number of transformer blocks (1–4), number of neurons (16–128), activation function (*'relu'*, *'tanh'*, *'sigmoid'*), optimizer (*'adam'*, *'sgd'*, *'rmsprop'*), learning rate (0.0001–0.01), decay (0.0001–0.01), batch size (16–128) and number of epochs (10–100). This systematic approach allowed us to thoroughly investigate the impact of each hyperparameter on the model's performance and ultimately identify the optimal combination for our specific task. The optimal combination of hyperparameters for the proposed models on different datasets are presented in Tables 3, 4 and 5, respectively using PSO, GWO and WOA algorithms.

Table 3 Optimal hyperparameters obtained for Transformer-GWO model

Hyperparameters	Transformer-GWO				
	Dehradun	Farrukhabad	Ludhiana	Pipli	Una
Head size	16	17	29	128	92
Number of heads	1	1	2	1	1
Feed-Forward dimension	127	20	62	18	100
Number of Transformer blocks	2	3	4	1	3
Number of neurons	128	128	82	128	89
Activation function	*relu*	*relu*	*relu*	*tanh*	*relu*
Optimizer	*adam*	*adam*	*adam*	*adam*	*adam*
Learning rate	0.006	0.005	0.006	0.007	0.004
Decay	0.002	0.004	0.007	0.002	0.001
Batch size	16	24	36	28	16
Number of epochs	100	100	91	87	100

Table 4 Optimal hyperparameters obtained for Transformer-WOA model

Hyperparameters	Transformer-WOA				
	Dehradun	**Farrukhabad**	**Ludhiana**	**Pipli**	**Una**
Head size	111	27	37	28	23
Number of heads	1	4	1	4	1
Feed-Forward dimension	30	25	20	115	101
Number of Transformer blocks	1	2	2	1	2
Number of neurons	128	124	128	128	128
Activation function	*tanh*	*tanh*	*relu*	*tanh*	*sigmoid*
Optimizer	*adam*	*adam*	*adam*	*adam*	*adam*
Learning rate	0.002	0.002	0.002	0.004	0.0008
Decay	0.006	0.01	0.01	0.009	0.007
Batch size	16	65	39	47	22
Number of epochs	100	77	100	100	100

Table 5 Optimal hyperparameters obtained for Transformer-PSO model

Hyperparameters	Transformer-PSO				
	Dehradun	**Farrukhabad**	**Ludhiana**	**Pipli**	**Una**
Head size	117	32	100	92	50
Number of heads	3	2	1	2	2
Feed-Forward dimension	83	30	103	58	51
Number of Transformer blocks	3	2	2	2	2
Number of neurons	69	97	99	92	100
Activation function	*relu*	*tanh*	*tanh*	*relu*	*relu*
Optimizer	*adam*	*adam*	*adam*	*adam*	*adam*
Learning rate	0.005	0.007	0.007	0.003	0.006
Decay	0.002	0.003	0.003	0.009	0.002
Batch size	94	75	104	21	74
Number of epochs	67	65	59	84	40

Table 6 Parameter settings for different standalone models

Model	Parameters
GARCH	GARCH (1,1); normal distribution; estimation via maximum likelihood
ANN	1 hidden layer, 64 neurons; activation: ReLU; optimizer: Adam; epochs: 100; batch size: 32
SVM	Kernel: Radial Basis Function; C = 10; gamma = 0.01
LSTM	1 hidden layer, 64 units; dropout: 0.2; optimizer: Adam; learning rate: 0.001; epochs: 50; batch size: 32
Transformer	Attention heads: 2; head size: 64; feed-forward dim: 64; blocks: 2; activation: ReLU; optimizer: Adam; learning rate: 0.001

Hyperparameters for all the other standalone models is provided in Table 6.

Sensitivity analysis

Each MHA hyperparameters significantly affect the quality of hyperparameter optimization. Sensitivity analysis was conducted by varying these settings and observing their impact on validation loss and convergence stability (Tables 7, 8 and 9).

Implementation of forecasting models

The performance of the proposed models was systematically evaluated against both traditional statistical models, specifically the Generalized Autoregressive Conditional Heteroscedasticity (GARCH) model and various machine learning models, including the Artificial Neural Network (ANN) and Support Vector Machine (SVM) [56–58]. Additionally, standalone DL models such as Long Short-Term Memory (LSTM) and Transformer were also considered for comparison. The optimization is done for the Transformer model because of its superior performance as a standalone model compared to other models. The results of the study unequivocally demonstrate that the proposed models consistently outperformed all of the aforementioned models, showcasing

Table 7 Sensitivity Analysis of GWO

GWO		
Parameter	**Values Tested**	**Observation**
Number of Wolves	5, 10, 20, 30	Higher wolf count improves solution diversity, but increases time per iteration
Coefficient Decay	Linear, adaptive	Adaptive decay stabilizes convergence in later iterations

Table 8 Sensitivity Analysis of WOA

WOA		
Parameter	**Values Tested**	**Observation**
Number of Whales	5, 10, 20	More agents enhance exploration but amplify oscillation near optima
Spiral Coefficient	1.5, 2.0, 2.5	Higher values increase the amplitude of search; risk of overshooting minima

10 whales with moderate spiral coefficients (1.5–2.0) yield better convergence without excessive oscillation.

Table 9 Sensitivity Analysis of PSO

PSO		
Parameter	**Values tested**	**Observation**
Number of Particles	10, 20, 30	Faster convergence with 20 + particles, but risk of premature stagnation
Inertia Weight	0.3, 0.5, 0.7	Lower inertia encourages exploitation; higher promotes exploration
Acceleration Coefficients (c_1, c_2)	(1.5,1.5), (2.0,2.0), (2.5,2.0)	Higher values lead to faster convergence but increase instability

superior forecasting performance. This superiority is elaborated upon in further detail in the subsequent discussion.

After fine-tuning, the aforementioned predictive models, namely Transformer-PSO, Transformer-GWO and Transformer-WOA, underwent training with optimized parameters. The training process incorporated an early stopping criterion to mitigate overfitting and stabilize the models for subsequent forecasting on test data. In a comparative analysis encompassing five diverse markets (Dehradun, Farrukhabad, Ludhiana, Pipli and Una), the performance of the three hybrid Transformer models was evaluated using key metrics, including RMSE and MAE.

For the Dehradun market, the Transformer-WOA model demonstrated the highest forecasting accuracy with the lowest RMSE of 63.77 and MAE of 44.69. The Transformer-PSO and Transformer-GWO models also exhibited strong performance with RMSE values of 64.98 and 72.55 and MAE values of 46.23 and 50.18, respectively. These results underscore the ability of metaheuristic-enhanced Transformer models to handle the complexity of the data. In contrast, the GARCH model showed the poorest performance with the highest RMSE of 614.24 and MAE of 512.15. This highlights its inability to capture the intricate patterns of this market. While ANN and SVM demonstrated moderate improvements over GARCH, they were significantly outperformed by the Transformer-based models. In addition, for the Farrukhabad market, the Transformer-GWO model outperformed the others, with the lowest RMSE of 35.78 and MAE of 28.27. Transformer-PSO followed closely with an RMSE of 36.46 and MAE of 28.31. Traditional models like GARCH and ANN showed significantly higher errors, with the GARCH model producing an RMSE of 77.49 and MAE of 62.78 and ANN showed marginal improvements. However, their performance was far behind the Transformer models, emphasizing the limitations of traditional forecasting methods. For the Ludhiana market, the Transformer-GWO model again delivered the best results with the lowest RMSE of 119.86 and MAE of 88.49. While the RMSE and MAE values were higher than in other markets like Farrukhabad, the Transformer-GWO still outperformed all other models in this region. On the other hand, the GARCH model again exhibited poor performance with an RMSE of 559.62 and MAE of 442.0, underscoring its limited effectiveness in this market. In Pipli, the Transformer-WOA model achieved the best results with an RMSE of 60.69 and MAE of 35.42. This demonstrates its superior accuracy in a market where other models struggled. For example, the GARCH model had the highest RMSE of 1372.61 and MAE of 1132.24, indicating much poorer predictive performance compared to the advanced models. Lastly, in Una, the Transformer-WOA model again showed the best results, with the lowest RMSE of 79.42 and MAE of 58.84. Transformer-PSO and Transformer-GWO also performed well, with RMSE values of 85.18 and 115.07, respectively. While the GARCH model continued to exhibit higher errors with an RMSE of 689.73 and MAE of 593.58.

Across the five markets considered under study, the metaheuristic-enhanced Transformer models consistently demonstrated superior percentage improvements in RMSE and MAE compared to the benchmark GARCH model in capturing the volatility (as shown in Table 10). Transformer-PSO exhibited the highest performance gains in Dehradun, with 89.42% improvement in RMSE and 90.97% in MAE, closely followed by Transformer-WOA with 89.61% and 91.27% improvements in RMSE and MAE,

Table 10 Percentage improvement (%) in the performance of different models considering GARCH as benchmark model

Models	Markets									
	Dehradun		Farrukhabad		Ludhiana		Pipli		Una	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
ANN	50.70	52.25	18.95	23.62	39.11	36.36	61.39	60.68	42.6	45.09
SVM	67.61	70.78	38.01	28.38	68.19	69.05	83.31	82.09	80.59	83.45
LSTM	64.46	67.87	39.87	35.09	71.62	61.97	87.96	82.77	82.48	83.81
Transformer	71.45	73.61	41.26	39.82	73.12	69.07	93.67	94.57	86.47	87.98
Transformer-PSO	89.42	90.97	52.94	54.90	75.05	76.87	95.28	96.36	87.65	89.18
Transformer-GWO	88.18	90.20	53.82	54.96	78.58	79.98	94.48	94.62	83.31	83.75
Transformer-WOA	89.61	91.27	48.45	47.14	73.26	76.49	95.57	96.87	88.48	90.08

respectively. Transformer-GWO also delivered strong results, with improvements of 72.55% in RMSE and 50.18% in MAE. In comparison, traditional models such as ANN and SVM displayed moderate improvements, with ANN achieving a 32.45% reduction in RMSE and SVM achieving 45.39%. The GARCH model, however, exhibited the least accuracy with significantly higher error values. Farrukhabad results indicated that Transformer-GWO and Transformer-PSO models outperformed others, with Transformer-GWO leading 53.82% in RMSE improvement and 54.96% in MAE improvement. Transformer-PSO followed closely, showing a 52.94% improvement in RMSE and 54.90% in MAE. Transformer-WOA demonstrated strong performance with an RMSE of 39.94 and MAE of 33.18, further highlighting its effectiveness in this market. Ludhiana saw the greatest percentage improvements from Transformer-GWO, which achieved 78.58% in RMSE and 79.98% in MAE. Transformer-PSO delivered 75.06% in RMSE and 76.88% in MAE compared to the GARCH model. Models such as ANN and SVM displayed relatively moderate performance gains, achieving 39.12% and 68.21% improvements in RMSE and 36.35% and 69.06% improvements in MAE, respectively. In the Pipli market, the Transformer-WOA model demonstrated the best performance, achieving the lowest RMSE of 60.69 and MAE of 35.42, corresponding to improvements of 95.57% in RMSE and 96.87% in MAE compared to the GARCH model. Transformer-PSO also exhibited strong performance, with an RMSE of 64.72 and MAE of 41.14, reflecting improvements of 95.29% in RMSE and 96.36% in MAE, respectively. In contrast, traditional models, such as ANN and SVM, displayed relatively modest improvements, with 61.39% and 83.31% improvements in RMSE and 60.70% and 82.08% in MAE, respectively. Una results showed Transformer-WOA leading in performance, with 88.48% improvement in RMSE and 90.08% in MAE compared to the GARCH model. Transformer-PSO and Transformer-GWO, also demonstrated strong predictive performance, showcasing their superior predictive accuracy in this market.

The robustness and generalizability of the Meta-Transformer models, particularly Transformer-WOA and Transformer-GWO, were evident across diverse market conditions, with the lowest RMSE and MAE values achieved in markets like Dehradun, Pipli and Una. The models consistently outperformed not only the traditional statistical methods but also machine learning and deep learning models, including LSTM and SVM, in error reduction and adaptability (Table 11). Additionally, the optimization

Table 11 Forecasting performance of all the models on different markets

Models	Dehradun		Farrukhabad		Ludhiana		Pipli		Una	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
GARCH	614.24	512.15	77.49	62.78	559.62	442.01	1372.61	1132.24	689.73	593.58
ANN	302.77	244.51	62.8	47.95	340.74	281.29	529.83	445.16	395.88	325.93
SVM	198.95	149.62	48.03	44.96	177.98	136.78	229.05	202.75	133.84	98.2
LSTM	218.24	164.53	46.59	40.75	158.79	168.09	165.16	195.02	120.78	96.06
Transformer	175.33	135.13	45.51	37.78	150.38	136.7	86.85	61.46	93.31	71.33
Transformer-PSO	64.98	46.23	36.46	28.31	139.62	102.2	64.72	41.14	85.18	64.22
Transformer-GWO	72.55	50.18	**35.78**	**28.27**	**119.86**	**88.49**	75.65	60.91	115.07	96.4
Transformer-WOA	**63.77**	**44.69**	39.94	33.18	149.59	103.88	**60.69**	**35.42**	**79.42**	**58.84**

Bold values indicate the best performing model

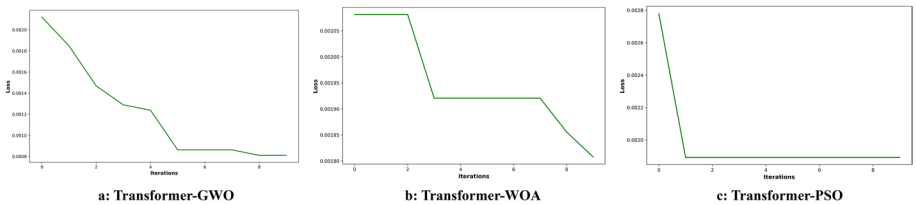

Fig. 6 Convergence curves of the Meta-Transformer models for the Dehradun market

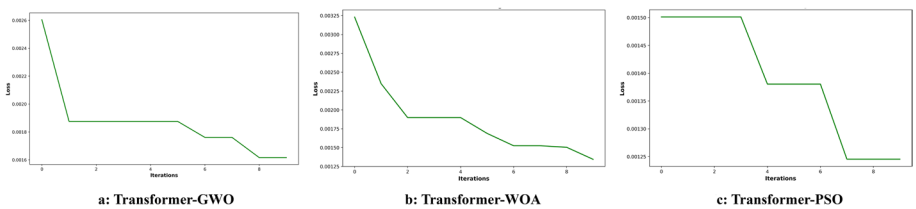

Fig. 7 Convergence curves of the Meta-Transformer models for Farrukhabad market

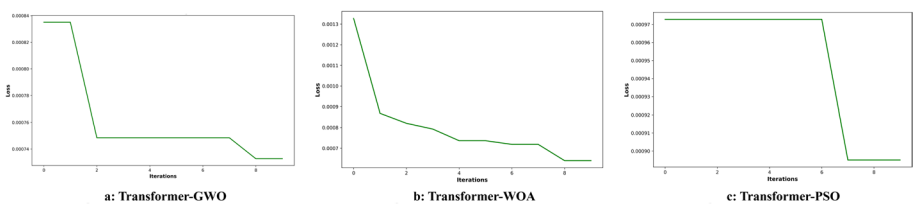

Fig. 8 Convergence curves of the Meta-Transformer models for Ludhiana market

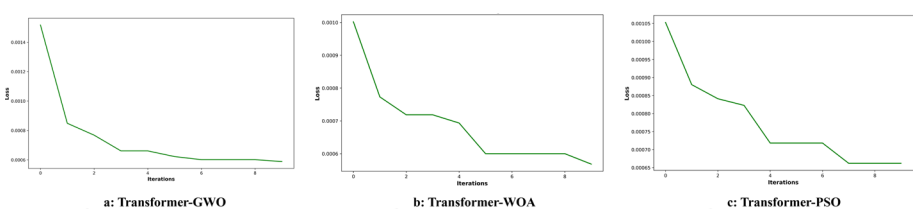

Fig. 9 Convergence curves of the Meta-Transformer models for Pipli market

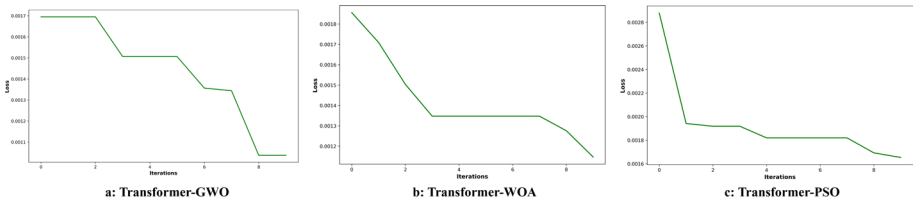

Fig. 10 Convergence curves of the Meta-Transformer models for Una market

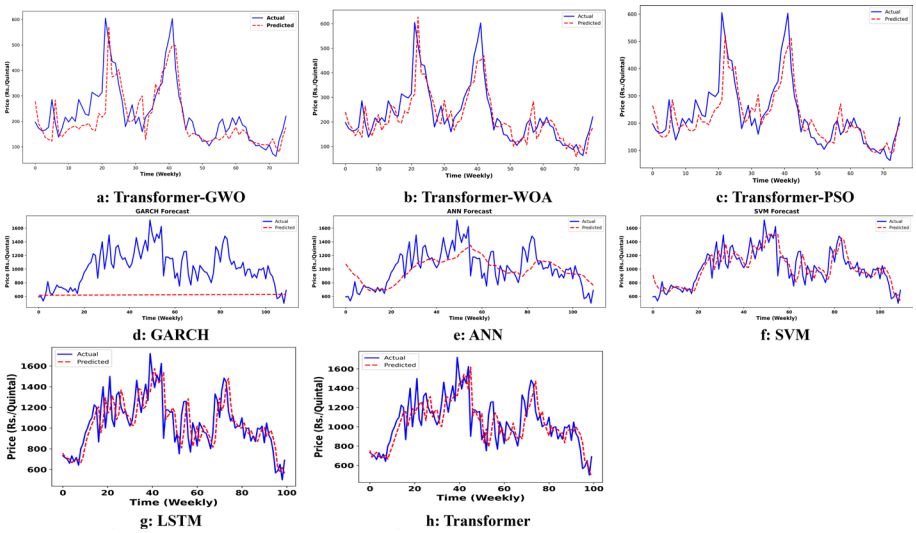

Fig. 11 Forecasting performance of different models for Dehradun market on test data

process within these models ensured a balanced trade-off between exploration (global optima) and exploitation (local optima), leading to fine-tuned hyperparameters and enhanced model performance. The computational efficiency of the models, attributed to the parallelization capabilities of the Transformer and the fast convergence of MHAs (shown in Figs. 6, 7, 8, 9, 10), makes them highly suitable for real-time agricultural price forecasting applications where timely predictions are essential. Figures 11, 12, 13, 14, 15 visually demonstrate that the forecasted price series is closely aligned with the actual price values, further highlighting the accuracy of the Meta-Transformer models in capturing price dynamics. The study indicates that PSO achieves rapid convergence but is prone to premature stagnation, potentially leading to suboptimal results; GWO offers steady and robust convergence, making it ideal for long-term training stability; while WOA demonstrates strong exploration ability but may suffer from instability in convergence. The Meta-Transformer models demonstrated a remarkable ability to handle non-linear data patterns and complex market dynamics, proving to be highly effective tools for decision-making in the agricultural sector. These findings offer a solid foundation for future studies to expand on this framework, applying it to other crops and markets and investigating potential combinations with advanced deep learning architectures like GRU, LSTM, NBEATS or hybrid models.

In future research, the proposed framework can be extended in several promising directions. Reinforcement learning (RL) could be integrated into the training

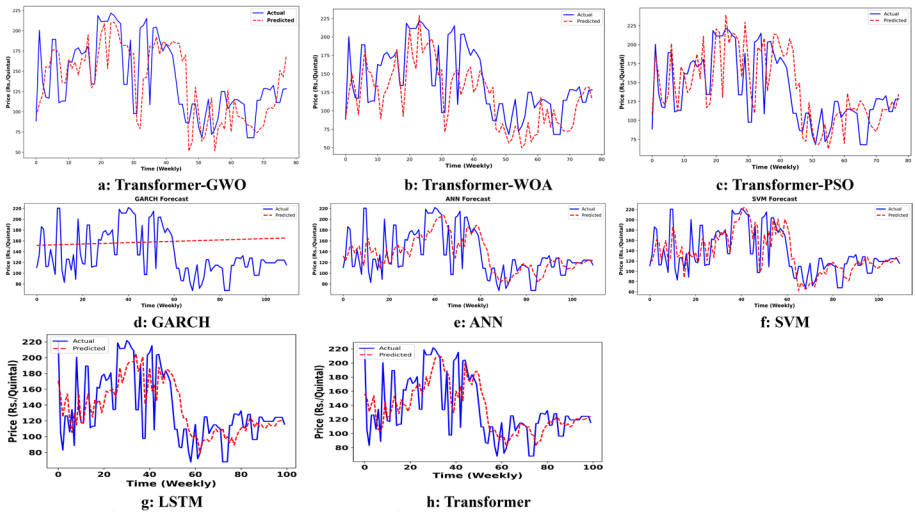

Fig. 12 Forecasting performance of different models for Farrukhabad market on test data

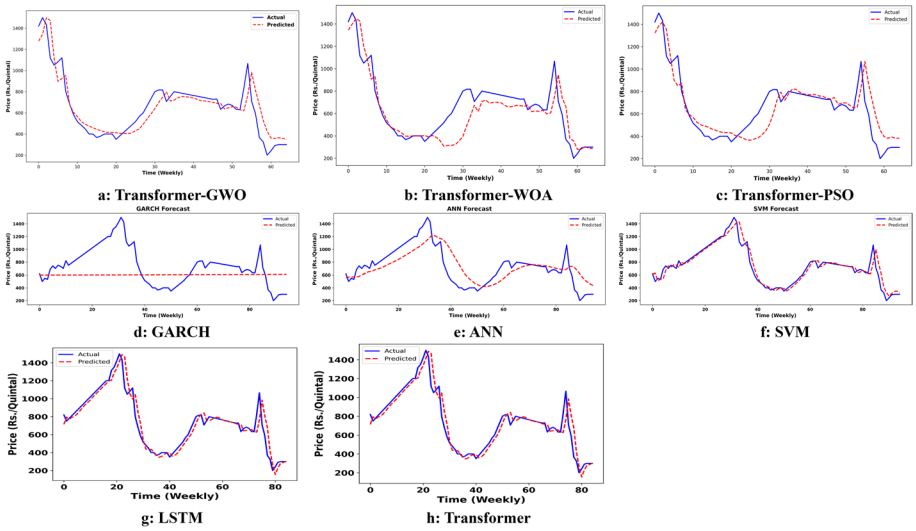

Fig. 13 Forecasting performance of different models for Ludhiana market on test data

loop to enable adaptive forecasting, where model hyperparameters or strategies are dynamically fine-tuned based on real-time feedback from market environments. This approach would be particularly valuable under conditions of high volatility or abrupt structural changes. Additionally, incorporating ensemble learning strategies, such as combining GRU, CNN-LSTM, or Transformer variants optimized with different metaheuristics, can help improve model robustness and generalization. These ensemble-based systems could capture complementary patterns across architectures, resulting in more stable and accurate forecasts. Lastly, it is important to consider the ethical dimensions of AI-based forecasting, particularly with respect to data fairness. Agricultural datasets may carry biases due to uneven data collection across regions or market segments. If left unaddressed, these biases could lead to inequitable

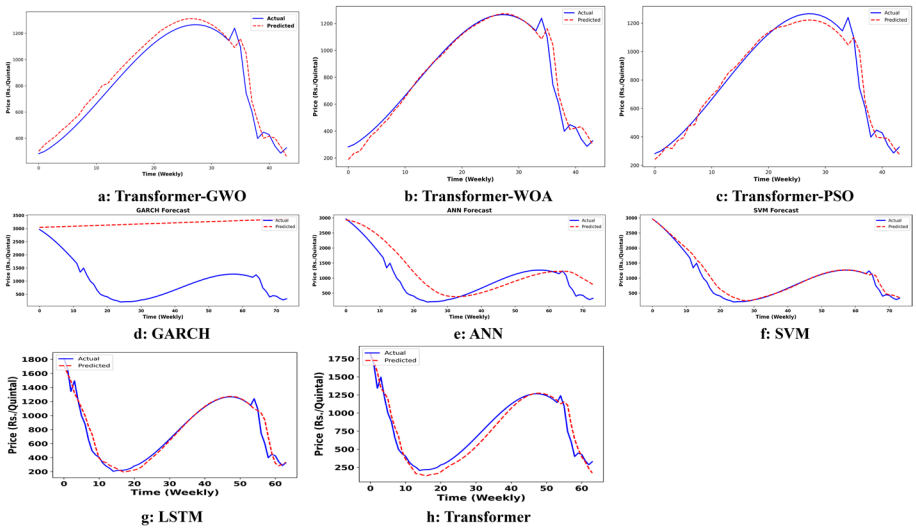

Fig. 14 Forecasting performance of different models for Pipli market on test data

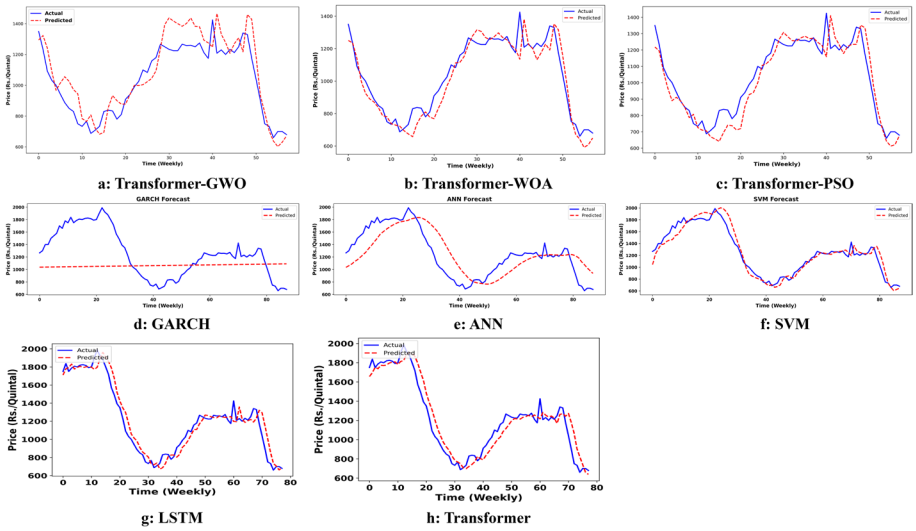

Fig. 15 Forecasting performance of different models for Una market on test data

recommendations, disproportionately impacting underrepresented farming communities. Future work should aim to include fairness-aware modeling techniques and emphasize transparent evaluation metrics to promote equitable outcomes in agricultural forecasting systems.

Conclusions

Agricultural commodity price forecasting plays a vital role in strengthening farmers' livelihoods, especially in a country like India, where agriculture is a key economic pillar. This study presents an innovative approach that integrates Transformer models with Metaheuristic Algorithms (MHAs) such as Grey Wolf Optimizer (GWO), Whale Optimization Algorithm (WOA), and Particle Swarm Optimization (PSO) to improve the

accuracy of agricultural price predictions. The Transformer architecture, known for its self-attention mechanism and parallel processing capabilities, is enhanced through these MHAs to ensure efficient training and faster convergence. Evaluated on weekly potato price data from five major markets in Northern India, the proposed Meta-Transformer models, particularly Transformer-GWO and Transformer-WOA, significantly outperformed traditional statistical, machine learning, and deep learning models, including the widely used GARCH model. The results showed a remarkable 70–90% improvement across key metrics such as RMSE and MAE, underscoring the potential of this hybrid framework in delivering accurate and timely price forecasts.

Beyond technical advancements, the real-world implications of this research are profound. Accurate forecasting tools can inform agricultural policies, aid in market stabilization, and provide farmers with actionable insights for decision-making related to crop planning, storage, and sales. However, certain limitations remain—this study focuses on a single crop and geographic region, and the model's performance is dependent on the availability and quality of historical price data. Additionally, the computational intensity of Transformer models optimized with MHAs may limit their deployment in low-resource environments. Future work could expand the scope to include multiple crops, integrate external factors like weather and economic indicators, and explore advanced techniques such as hybrid models, reinforcement learning, and fairness-aware AI systems. This study lays a strong foundation for developing scalable, real-time, and equitable forecasting tools that can benefit stakeholders across the agricultural value chain.

Appendix

See Fig. 16

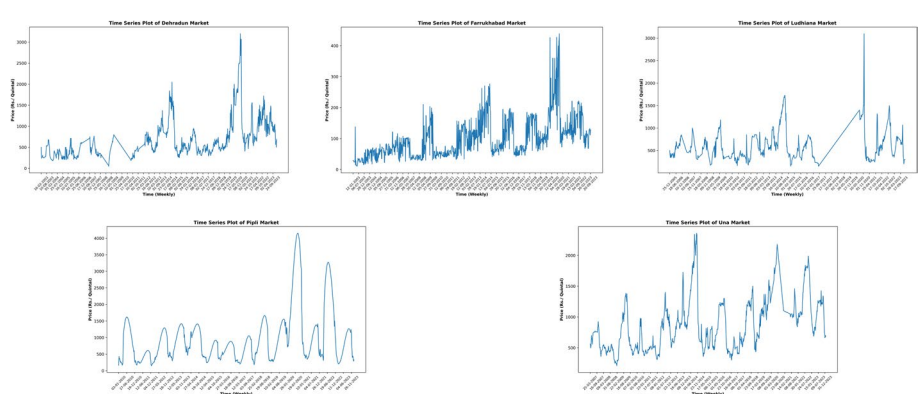

Fig. 16 Time series plots of different potato markets

Author contributions
All authors were involved in the conception and design of the study. Harish Nayak G. H., Wasi Alam, B. Samuel Naik, G. Avinash, Varshini B S, Mrinmoy Ray, and Rajeev Ranjan Kumar carried out material preparation, data collection, and analysis. Harish Nayak G. H., G. Avinash, B. Samuel Naik, Mrinmoy Ray and Rajeev Ranjan Kumar wrote the initial draft of the manuscript, with contributions from Wasi Alam to enhance its quality. K N Singh and Wasi Alam reviewed the manuscript. Every author reviewed and approved the final version.

Funding
The author(s) received no financial support for the research, authorship and/or publication of this article.

Availability of data and materials
The data can be accessed through "AgMarknet" website.

Declarations

Ethics approval and consent to participate
Not applicable.

Competing interests
The authors declare no competing interests.

Received: 11 December 2024 Accepted: 12 May 2025
Published online: 31 May 2025

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